

RIVERSIDE GENERAL HOSPITAL

Hospital Performance Analysis__ Power BI Project
Financial, Operational & Clinical Insights | Jan – Jun 2026

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Prepared as a data analytics case study using a Power Bi star schema Model

Operational & Clinical Insights Dashboard

Total Visits

Emergency Admission Rate

Average Length of stay

Recovery Rate

Average Satisfaction Score

3.9

RIVERSIDE GENERAL HOSPITAL: Revenue Overview

Department Name
All

Total Revenue
318K

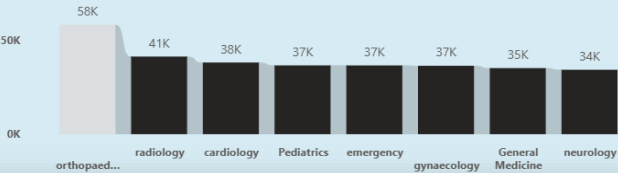
Total Patients
117

Collection Rate
76.2%

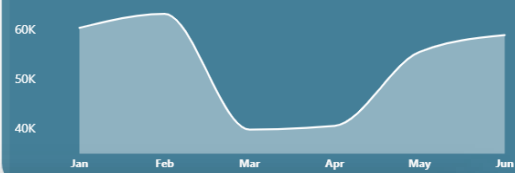
Total Outstanding Balance
24K

Average Wait Time
29.8

Revenue by Department



MOM Revenue



DepartmentName	Sum of Total Revenue
cardiology	38,266.31
emergency	36,696.98
gynaecology	36,509.28
orthopaedics	58,414.22
General Medicine	35,277.36
neurology	34,393.58
Pediatrics	36,719.05
radiology	41,442.98
Total	317,719.76

2

Avg Length of Stay

28.8%

Emergency Admission Rate

39.2%

Recovery Rate

318K

Total Revenue

480

Total Visits

117

Total Patients

24K

Total Outstanding Balance

29.8

Avg Wait Time

3.9

Avg Satisfaction Score

76.4K

Total Amount Paid

342K

Total Treatment Cost

241.3K

Total insurance Coverage

76%

Collection Rate

Summary

What this project is about

This project is a Power Bi analytics dashboard built for Riverside General Hospital, designed to give hospital management a single, consolidated view of how the organization is performing financially, operationally, and clinically.

AT A GLANCE

Dashboard: a 3-page Power Bi report which consists of KPIs, Revenue overview and an Operational & Clinical insights page built from a library of custom Dax measures.

Data Foundation: a star-schema model with four dimensions tables (Departments, Doctor, Patients, Procedures) connected to a central hospital visits fact table, covering two quarters (Jan-Jun 2026).

Purpose: to know the money we are owed and turn scattered visits level records into KPIs and visuals that answer real hospital management questions; Are we collecting what we bill? Where are patients waiting too long? Which department and doctors need attention.

Introduction

Understanding the type of project

This is a Business Intelligence (BI) / Data analytics project, built end-to-end in Microsoft Power Bi. It demonstrates the full analytics workflow from raw data to a decision-ready dashboard.

PROJECT STAGES COVERED

- **Data Modelling: structuring** raw hospital records into a relational star schema (facts + dimension tables) so the data can be sliced by department, doctor, patient and time.
- **Dax Measure Development:** It's the layer that turns the raw hospital data into reusable, trustworthy KPIs which respond to whatever the dashboard is filtered to.
- **Dashboard design:** translating those measures into KPI cards, charts and slicers organized around a clear visual hierarchy and a consistent color theme.
- **Insight generation: interpreting** the numbers to surface pattern hospital leadership can act on. Not just displaying data but explaining what it means.

Problem Statement

The problem the hospital needs to solve

Before this dashboard, hospital data on revenue, patient flow and clinical outcomes existed only as raw, disconnected visits record. Leadership had no fast way to see which departments were profitable, where money was going uncollected, or where patients were experiencing delays, all of which affects finance and care quality.

Key Questions the Dashboard Needs to Answer

- Which departments generate the most revenue, and which are underperforming?
- How much of what patient owed has been collected and how much is at risk?
- Are patients waiting too long for care, and does that affect satisfaction or outcome?
- What is our mix of emergency vs. Scheduled visits, and are we resourced for it?
- Which regions and doctors should leadership focus attention or support on?

Data Overview

What data is about

The model follows a star schema: one central fact table of hospital visits, linked to four supporting tables and a standard date table.

The image displays three screenshots of a database management tool, likely Microsoft SQL Server Enterprise Manager, showing the structure and data of a hospital database. The first screenshot shows the 'Fact Hospital Visits' table, which is the central fact table in the star schema. It contains 480 rows of data, representing hospital visits. The second screenshot shows the 'Dim Departments' table, which is a dimension table containing 8 rows of data, representing different hospital departments. The third screenshot shows the 'Dim Doctors' table, which is another dimension table containing 24 rows of data, representing different doctors. The tables are linked in a star schema, with the fact table connected to the dimension tables and a standard date table.

Table Name	Row Count	Columns
Fact Hospital Visits	480	VisitID, PatientID, DoctorID, DepartmentID, VisitDate, VisitTime, VisitDuration, VisitCost, VisitStatus
Dim Departments	8	DepartmentID, DepartmentName, Floor, BedCapacity, StaffCount
Dim Doctors	24	DoctorID, DoctorName, DepartmentID, Specialty, Grade, YearsExperience

Hospital Visits (Fact): (480 rows) Total of 6 months Jan- Jun. Suggesting each month contributed roughly 80 visits. It is the core transactional record of every visit.

Dim_Departments: (8 rows) Department ID, Department name, floor, Bed capacity, staff count. The department table helped with capacity and staffing analysis.

Dim_Doctors: (24 rows) Doctor ID, DoctorName, DepartmentID, Specialty, Grade, YearsExperience. The doctors roster used to track performance.

Dim_Patients: (120 rows) PatientID, Firstname, Lastname, DateOfBirth, Gender, City, BloodGroup, InsuranceProvider. The patient demographics were used for population level analysis.

Dim_Procedures: (18 rows) DepartmentID, ProcedureID, ProcedureName, Standardcost. The procedure catalogue was used to benchmark actual billed cost against standard cost.

Date Table: Date, Day Number, Day of Week, Month Name/Number, Quarter, Year. Enables month-over-month and quarterly trend analysis.

Data Preparation & Summary

Data cleaning process in power query

Before any measure or visual was built, the raw visit file was combined and cleaned in Power query. The applied steps pane shows the exact pipeline used to turn scattered sources file into a reliable, analysis ready table.

Combine & Shape: monthly visits files are merged into one consistent table with the right columns before any analysis begins.

Fix Data Types: Dates, numbers and text are set to the right data type, so DAX calculations and time intelligence work correctly.

Text Standardization: fixes the exact casing/whitespace problem flagged vales like “Paid”, “PAID” and “paid” are unified into one consistent category, so filters and measures count them correctly.

Two Cleaning Passes: applied separately to different categorical columns (e.g. AdmissionType/Outcome/PaymentStatus, then Gender) since each needed the same trim and case fix.

Remove Duplicates: prevents the same visit from being double counted in totals like Total Visits or Total Revenue.

Created calculated columns: three columns were added directly in power query (rather than DAX measures), so they are available row-by-row for slicing, filtering & grouping throughout the model.

- **Total Revenue:** combines what the patient paid and what insurance covered into one true revenue figure per visit. separate from the billed treatment cost.

Custom Column

Add a column that is computed from the other columns.

New column name
Total Revenue

Custom column formula
= [InsuranceCoverage] + [AmountPaid]

Available columns
LengthOfStayDays
WaitTimeMinutes
TreatmentCost
InsuranceCoverage
AmountPaid
PatientSatisfactionScore
Outcome

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

- **Outstanding Balance:** the unpaid remainder per visit after insurance and patient payments. Rolls up into the outstanding balance of KPI.

Custom Column

Add a column that is computed from the other columns.

New column name
Outstanding Balance

Custom column formula
= [TreatmentCost] - [Total Revenue]

Available columns
AmountPaid
PatientSatisfactionScore
Outcome
PaymentStatus
Total Revenue
Outstanding Balance
WaitTimeBand

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

WaitTimeBand: (conditional column) Buckets each visit wait time into four readable bands, used across the wait time and satisfaction charts instead of a raw minute value.

Add Conditional Column

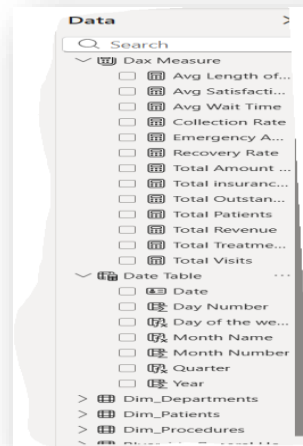
Add a conditional column that is computed from the other columns or values.

New column name: WaitTimeBand

Column Name	Operator	Value	Output
WaitTimeMinutes	is less than or equal to	15	Short Time
WaitTimeMinutes	is less than or equal to	30	Moderate Time
WaitTimeMinutes	is less than or equal to	60	Long Time
Add Clause			
Else			Null

OK Cancel

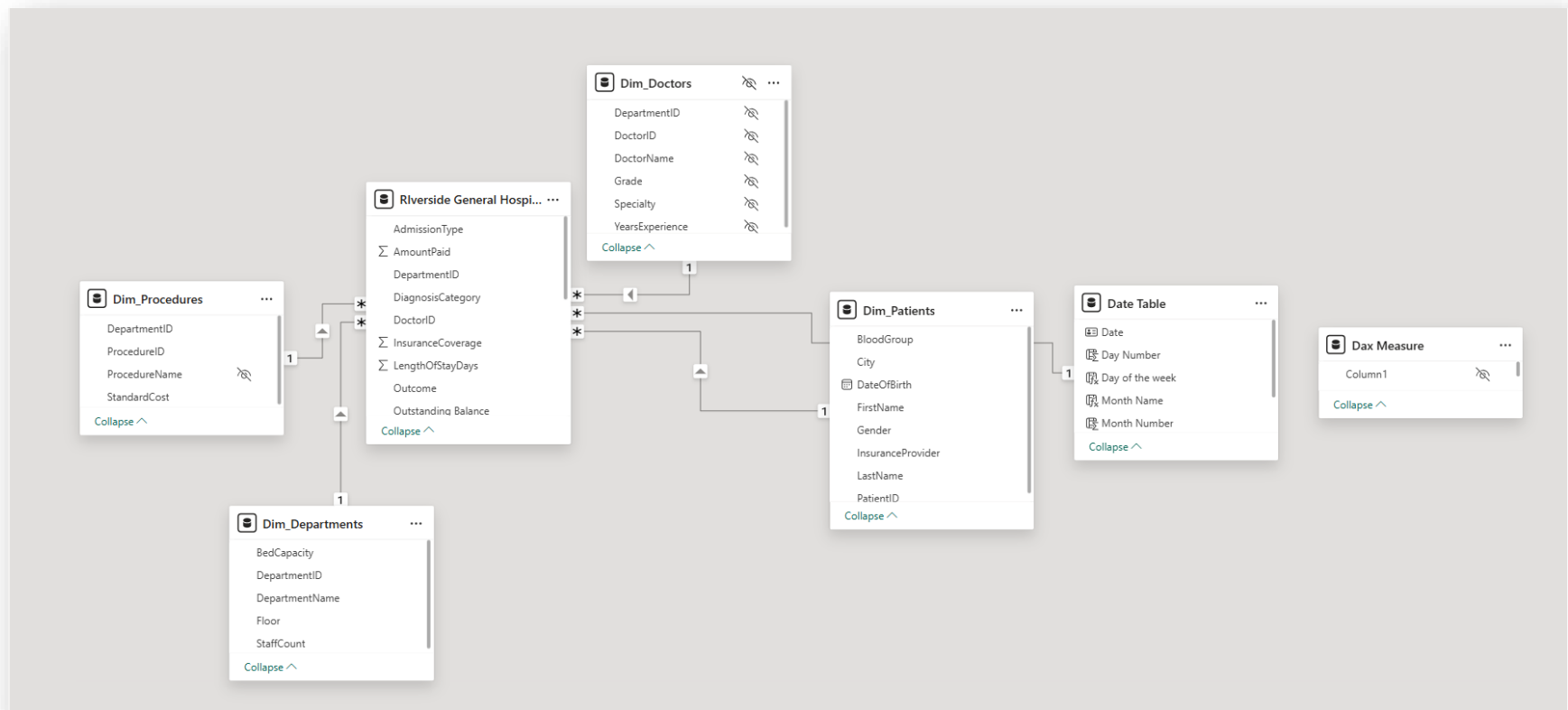
DAX Measures: calculated 13 DAX measures needed for analysis, which are grouped into Financial, Volume, Operational and Clinical experience each with its formula and logic.



Relationship Model

How the tables connect

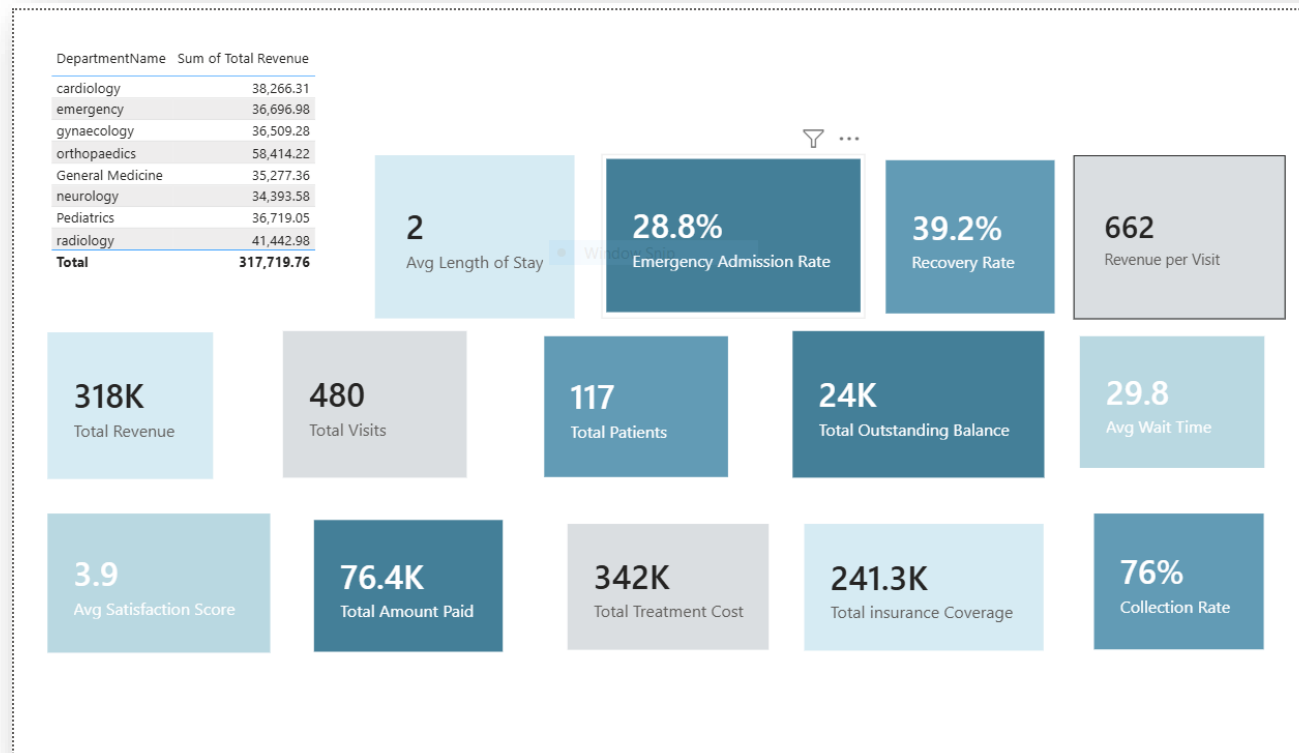
The model is built as a star schema; one central fact table sits in the middle, and every dimension table connects it directly through a one-to-many relationship on a shared ID key. No dimension table connects to each other directly; every relationship routes through the fact table.



Key Findings

KPI snapshot across the dashboard

Before breaking down each page, here are the Key Performance Indicators.



Total Revenue: (combined patients payments + insurer payments across two quarters 318k) reflects what's actually collected, not just billed, the financial performance indicator.

Collection Rate: (76.2% of a quarter of what's owed after insurance is still uncollected) it directly explains the 24k outstanding balance, a moderate, not critical, collections gap.

Total Visits: (480 steady volume over Jan-Jun) baseline for measuring workload and capacity planning across all departments.

Total Outstanding Balance: (24k of real money earned through care but not yet recieved) flags cashflow risk; the natural target for collections follows up.

Total Patients: (total of 117, 4 visits per patient on average) signals a repeat visit rather than mostly one-off cases, relevant for care continuity planning.

Emergency Admission Rate: (28.8% of roughly 3 in 10 visits are unplanned) it is quite normal to moderate load for a general hospital, which confirms most visits are scheduled, aiding staffing predictability.

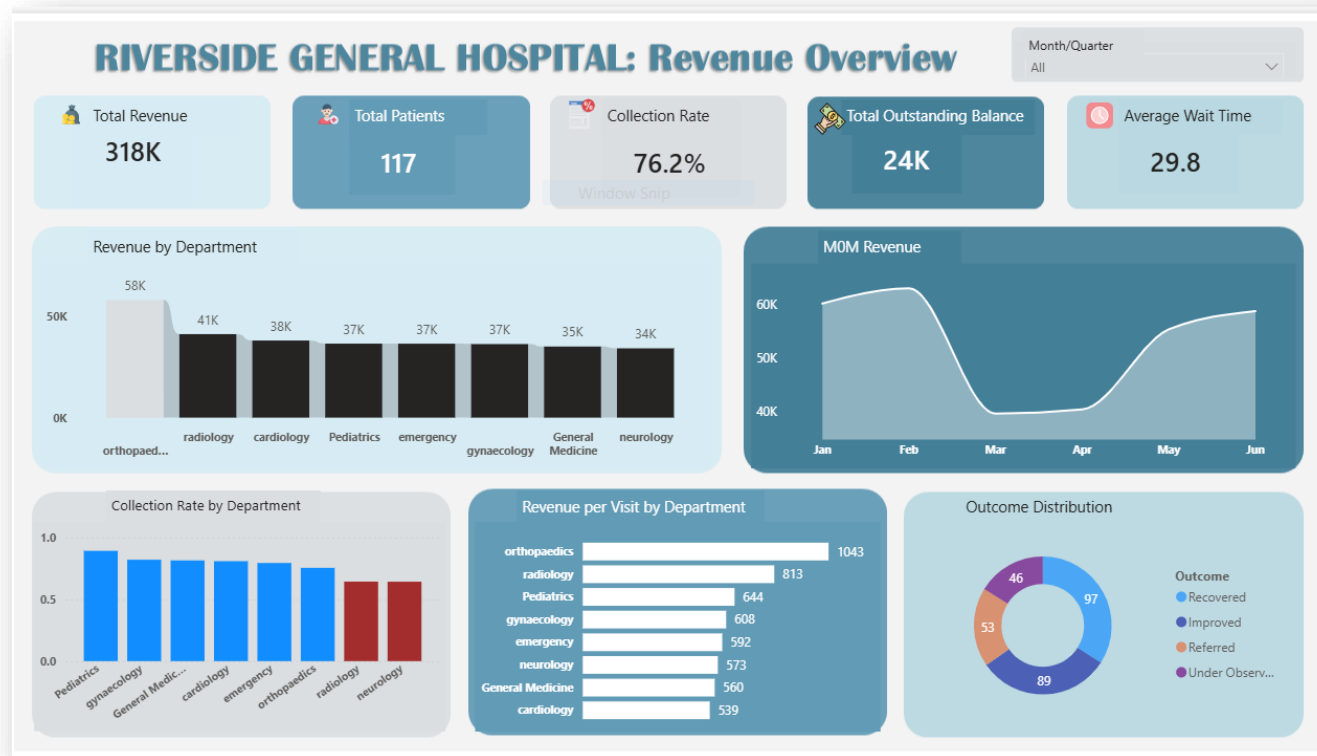
Recovery Rate: (39.2% of visits end specifically in "Recovered") not alarming on its own, but worth watching overtime rather than in isolation.

Analysis Questions and Insights

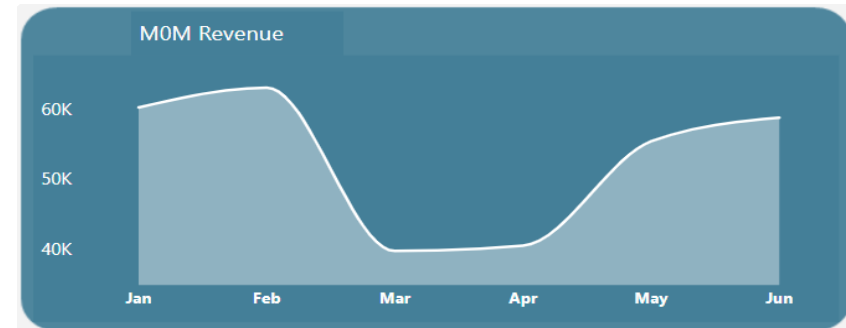
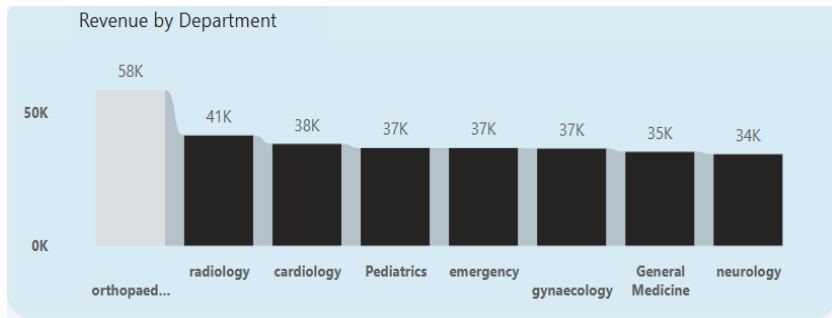
The report is built as two pages

Revenue Overview, Operational and Clinical Insights dashboard, each answering a different set of management questions using standard Power BI visuals.

Revenue Overview: "Are we earning what we should, and collecting what we're owed?"



- **Revenue by Department & M0M Trend**

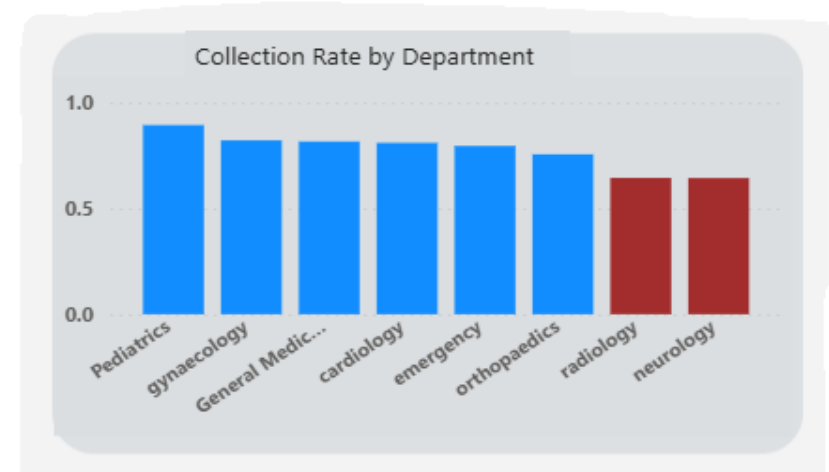


What department generates most revenue & which month during the quarter did we register most revenue?

- Orthopedics dominates at 58k, which is about 40% more than the next highest department. It displays a concentrated revenue base, not an evenly spread one.
- Revenue holds around 55-60k in Jan, drops sharply through February - March (bottoming near 40k), then recovers by May-June back to 60k. Shows a real deep worth investigating, not much noise.

- **Collection Rate by Department**

Department	Collection Rate	Hospital Avg (76.2%)	Status
Pediatrics	89%	+12.8pts	Strongest collector
Gynaecology	82%	+5.8pts	Above average
General Medicine	82%	+5.8pts	Above average
Cardiology	81%	+4.8pts	Above average
Emergency	79%	+2.8pts	Above average
Orthopedics	76%	-0.2pts	Right at par
Radiology	64%	-12.2pts	Below average
Neurology	64%	-12.2pts	Below average



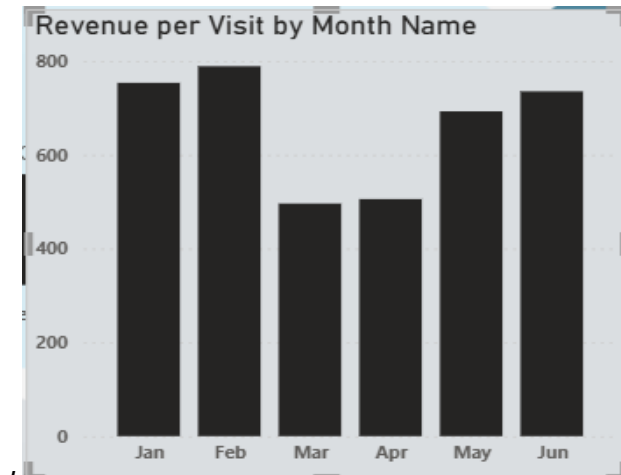
"Which departments are struggling to actually collect the revenue they've earned, and how far behind are they?"

At a glance, six **departments (Pediatrics through orthopedics)** sit at or above the hospital's **76.2% collection benchmark**, shown in blue, while **radiology and neurology** fall below it, shown in red — both trailing around 0.64-0.64, well behind the rest.

This turns a flat ranking into an instant decision tool: instead of scanning eight numbers and doing the comparison mentally, leadership can see in one glance exactly which two departments need a collections follow-up. Pediatrics stands out as the strongest collector (0.89), which is worth understanding *why*; its billing or follow-up process may hold lessons transferable to radiology and neurology.

- **Revenue per Visit by Department**

Department	Total Revenue	Revenue per visit	Visit by Department	Hospital Avg (662)
Orthopedics	58k	1043	71	+381
Radiology	41k	813	63	+151
Pediatrics	37k	644	62	-18
Gynecology	37k	608	60	-54
Emergency	37k	592	60	-70
Neurology	34k	573	57	-89
General Medicine	35k	560	56	-102
Cardiology	38k	539	51	-123



"Which departments generate the most value per patient encounter not just the most total revenue?"

- Orthopedics is the strongest department, generating 1,043 revenue per visit, 381 above the hospital average of 662. It also has the highest total revenue at 58k and the highest visit count at 71.
- Radiology ranks second, generating 813 per visit, 151 above the hospital average, with 41k total revenue and 63 visits.
- Six of the eight departments perform below the hospital average. Pediatrics is closest to the benchmark at 644, while Cardiology is lowest at 539.
- Cardiology generates 38k in total revenue, but its 539 revenue per visit is the lowest. This shows why total revenue alone does not measure revenue efficiency.

- Patient volume does not automatically translate into higher revenue per visit. The differences suggest service mix, procedures, pricing, and patient cases may influence revenue per encounter.
- Monthly revenue per visit fluctuates considerably. February is highest at about 780, followed by January at about 750 and June at about 730. March and April are the weakest months, at roughly 490 and 500.
- The March and April decline requires further investigation to determine whether it resulted from changes in department mix, patient spending, or the volume of higher-value services.

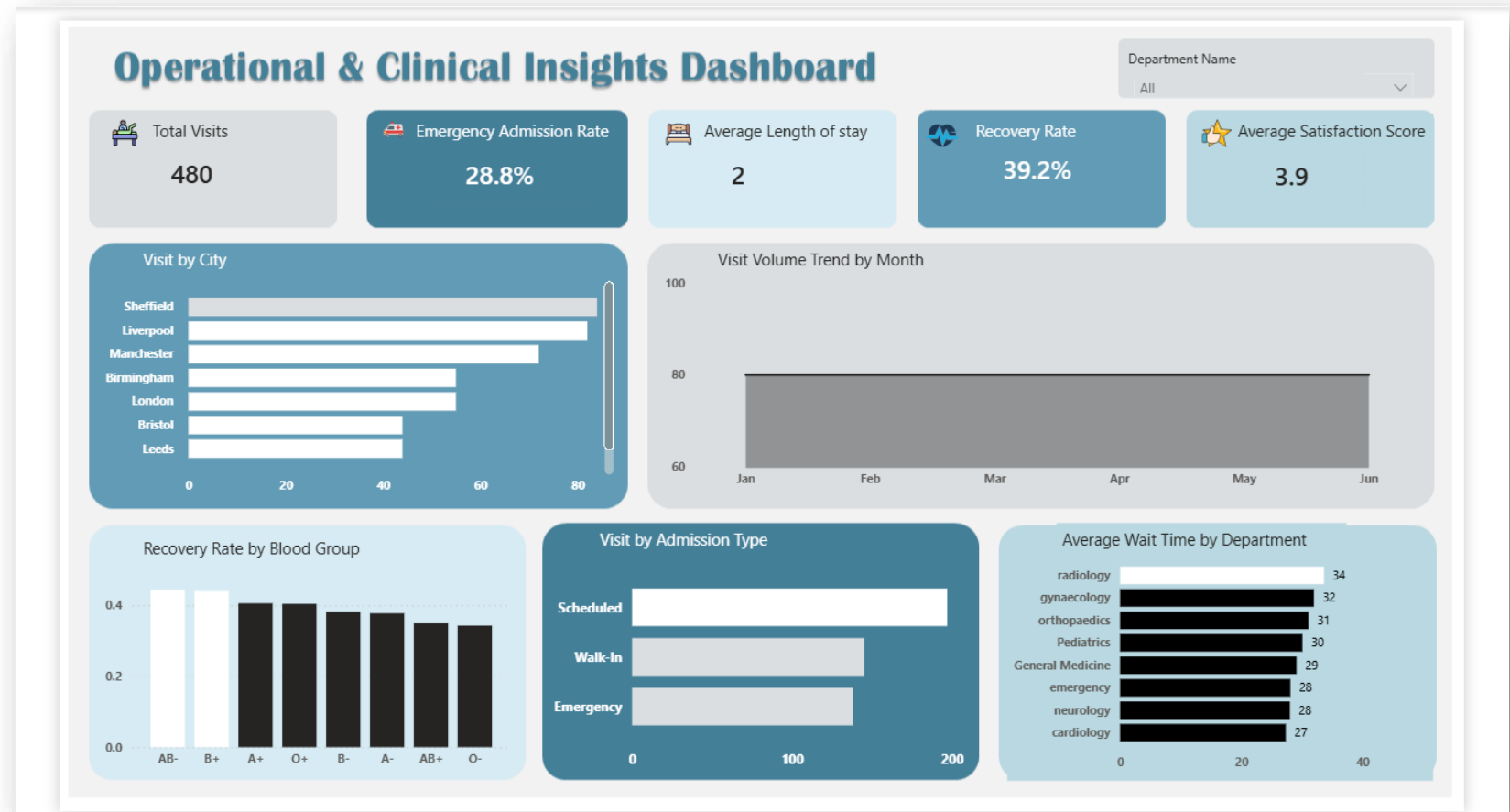
Overall, Orthopedics and Radiology are the only departments above the hospital average, while the results highlight the importance of measuring revenue per patient encounter alongside total revenue.

- **Outcome Distribution__Analysis**

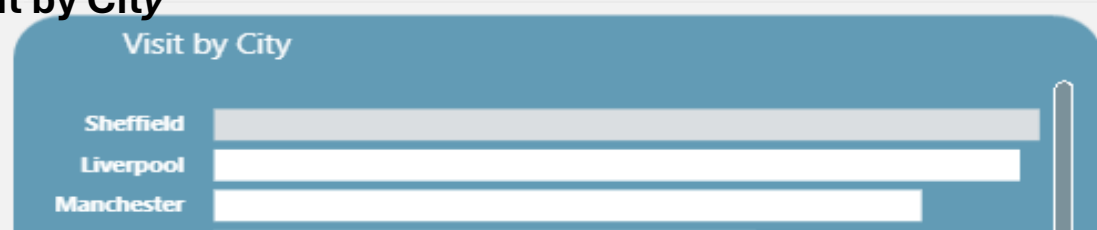
Outcome	Visits Count	% Share of visits
Recovered	188	39.2%
Improved	177	36.9%
Referred	61	12.7%
Under observation	54	11.25%

- Recovered + Improved = 365 visits (76%) __ the large majority of visits end in a positive clinical direction, either fully resolved or trending better. 39.2% (188) visits only captures full recovery, not an overall positive trajectory.
- Referred + Under Observation = 115 visits (24%) _roughly 1 in 4 visits require further care beyond that single visit, either escalated to a specialist (Referred) or kept for monitoring (Under observation). I believe this is a normal share for general hospitals, not a red flag on its own, but it's the group worth tracking overtime since a rising trend here would signal cases are not being resolved efficiently.

Operational and Clinical Overview: "How efficiently are we delivering care, and what outcomes and experience are patients getting as a result?"



- Visit by City



What region should leadership focus attention or support on?

Sheffield and Liverpool are two leading catchment cities with a singular visit of (84 & 82), Collectively accounting for large share of (166 visits) volume, which makes them the natural priority for local outreach, satellite clinics, or resource planning.

Bristol and Leeds trail well behind, roughly half the volume of Sheffield, worth investigating whether that's genuinely lower demand in those areas or an access/ awareness gap worth addressing.

- **M0M Trend (Visit Volume & Revenue) Comparison**

Month	Visit Volume	Revenue	Est.Revenue per visit
Jan	80	60k	750
Feb	80	63k	788
Mar	80	39k	488
Apr	80	40k	500
May	80	55k	688
Jun	80	58k	725

"Is our revenue performance driven by how many patients we're seeing, or by something else entirely?"

Visit volume is perfectly flat at exactly 80 visits every single month, while revenue swings from (39K to 58K) to a 50% range.

Revenue per visit dropped sharply — from 725 in January to ~500 in the March through, a real 30% decline per visit — then recovered back to 750 by June.

"Does patient volume explain the revenue swings we're seeing month to month?"

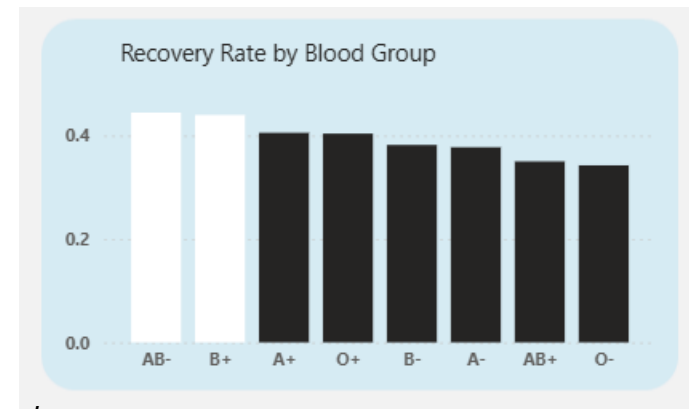
No visit volume does not explain it. Visit count stayed exactly flat at 80/month for all six months, while revenue swung by roughly 50% (39K to 58K) over that same period. Since the number of patients coming in never changed, patient demand can be ruled out as the cause of the Feb–March dip

This points to a **change in procedure mix, payer mix, or pricing** during Feb–March specifically, not a demand problem.

- **Recovery Rate by Blood Group**

Blood Group	Total Patients
AB-	17
A+	16
O-	16
AB+	15
A-	15
O+	14
B+	14
B-	13

Is there a particular blood type that isn't responding to treatment

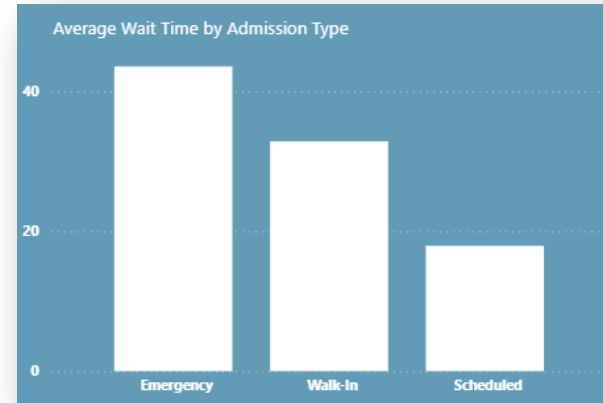
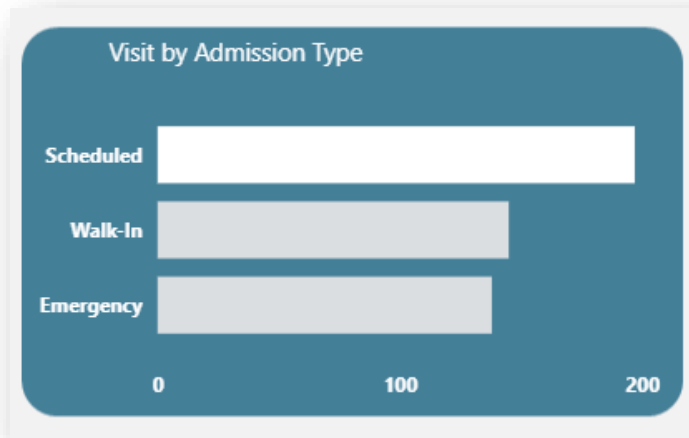


The spread across blood group is narrow (0.33 to 0.44), roughly an 11-point range, which is reassuring equity signal. No blood group is dramatically under or over performing recovery outcomes.

One caveat worth flagging: **AB- and B+ show the highest rates, and both are typically the rarest blood types in a general population** (AB- especially, often under 1%). "I believe their **higher** recovery rate could be a **small-sample artifact**, a handful of good outcomes pulling the average up rather than a genuine clinical pattern".

The counts are **fairly even across all eight groups** (13 to 17 patients each; a narrow spread). So, the recovery rate differences you're seeing aren't a small-sample fluke; each group has a reasonably comparable base to be measured against.

- **Visit by Admission Type & Average Wait Time by Admission Type**



Admission Type	Visit Volume	Avg Wait Time
Scheduled	200 (highest)	18 (shortest)
Walk-In	145	33
Emergency	140 (lowest)	42 (longest)

Scheduled visits (~200) are roughly 40% larger than either Walk-In or Emergency alone (~140-145 each), but Walk-In and Emergency are nearly identical in size to each other. That's the more interesting detail here: I'd have expected Emergency to be the smallest category, but it's carrying the same weight as Walk-In

Operational read: unplanned visits (Walk-In + Emergency combined, 285) are actually **comparable in total volume to Scheduled visits alone**. This means the hospital can't treat unplanned care as a small buffer around a mostly scheduled workload, nearly half of all visits arrive without a booked appointment, which

has real staffing implications: reception, triage, and department capacity all need to flex for a volume of unplanned demand that's almost as large as the planned side. This tells you the hospital needs standing capacity for unplanned care every single month, not just buffer staffing around a mostly scheduled base.

Which brings me to the table **pattern that appears inverse and not proportional**. The category with the *most* patients (Scheduled) has the *shortest* wait, while the category with the *fewest* patients (Emergency) has the *longest* wait. If wait time were simply driven by volume/congestion, you'd expect the opposite.

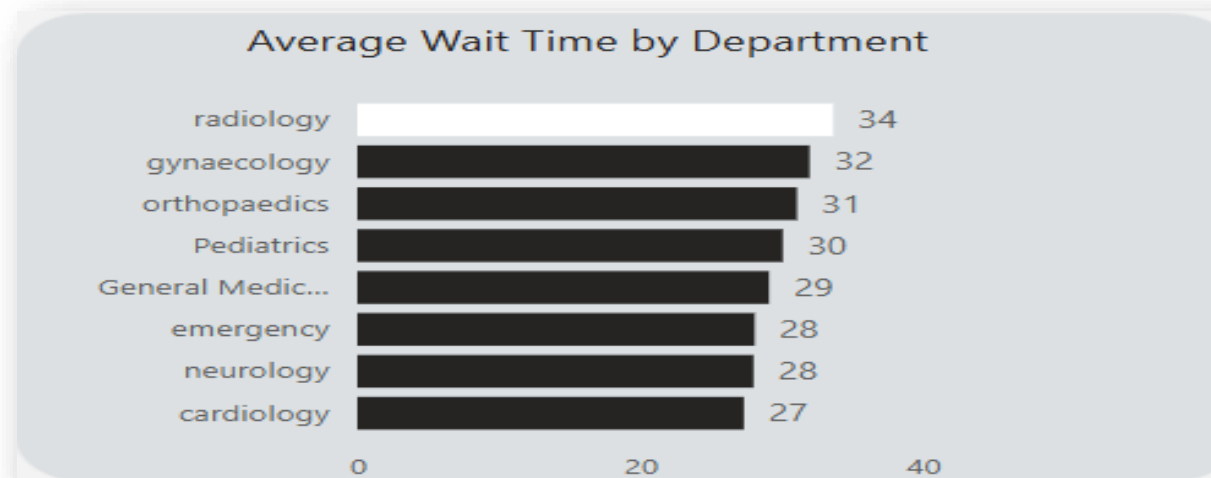
Why this makes operational sense: Scheduled visits have a reserved time slot — the provider and room are already allocated before the patient arrives, so there's minimal queuing regardless of how many scheduled visits happen that month. Walk-In and Emergency patients arrive unpredictably with no reserved capacity, so they queue for whatever's available on arrival — volume isn't the driver; **the scheduling mechanism itself is**.

The finding that actually needs flagging

Emergency having the longest wait of all three categories (42 min) is a genuine concern. It contradicts the reasonable assumption that emergency care is fast-tracked. A few possible explanations worth investigating.

- WaitTimeMinutes may measure check-in-to-first-contact time, and emergency intake (triage, vitals, initial assessment) genuinely takes longer to *record* even when clinically prioritized correctly.
- Emergency cases may be getting bottlenecked by the same unplanned-arrival problem as Walk-In, just worse, since severity varies more.

- **Average Wait Time by Department**



Are patients waiting too long, and does that affect satisfaction or outcomes?

- Radiology and Gynecology have the longest wait by department wide, but the spread across department is quite narrow) just 27-34 minutes, a 7-minute range. Compared to an average time of (29.8 min, hence KPI card), which is meaningful but not a crisis level issue.
- Emergency running below average (28 min) still holds up as expected. Triage prioritization is likely doing its job, keeping urgent cases moving faster than routine departments.
- Cardiology is genuinely the fastest at (27 min), worth understanding why; it may hold process lessons (staffing ratio, scheduling approach) that could transfer to Radiology & Gynecology

Recommendations

- Revenue & Collections

1. Focus collections follow-up on radiology and neurology first.

Both collect only ~63% of what patients owe, about 13 points below the 76.2% hospital average. The other six departments sit between 76% and 90%, so the gap is a sharp drop for two departments, not a gradual slope. Radiology is also a high-value department (813 per visit), so every uncollected radiology balance costs more than one elsewhere. The hospital-wide 24K outstanding balance is best recovered by targeting these two.

2. Protect orthopedics and radiology as the revenue engines.

Orthopedics earn 58K in total and 1,043 per visit, roughly double cardiology's 539 per visit. Radiology is a clear second tier at 813. The other departments cluster tightly between about 540 and 650 per visit. Any drop in orthopedic or radiology activity would move hospital revenue far more than the same drop elsewhere, so their capacity and staffing should be monitored closely.

3. Investigate the February to March revenue dip as a per-visit problem, not a demand problem.

Visits stayed flat at 80 per month while revenue fell from about 58K to 40K, and revenue per visit dropped roughly 30% (about 750 to 500) before recovering by June. Fewer patients cannot explain this. The likely causes are procedure mix, payer mix, or pricing.

- Operations and Patient experience

4. Reduce Emergency and Walk-In wait times before department-level waits.

Average wait by admission type is the sharpest operational finding: Scheduled 18 min, Walk-In 33 min, Emergency 42 min. Unplanned visits (Walk-In plus Emergency, 285 of 480) make up about 59% of all visits, so the longest waits fall on the majority of patients. Scheduled visits have reserved capacity. Unplanned arrivals do not, and the hospital carries this load every month with no seasonal relief. Options include a dedicated fast-track lane, flexible staffing at peak arrival times, and better triage.

5. Treat department wait times as a secondary, modest improvement.

Department averages range only from 27 minutes (cardiology) to 34 (radiology), against a 29.8 average. There is no severe outlier. Bringing radiology and gynaecology down by 3 to 4 minutes would be a realistic target, and cardiology's process is worth studying as a benchmark.

6. Plan local outreach around Sheffield and Liverpool.

Sheffield (84 visits) and Liverpool (82) are the top catchment cities, with Manchester third. Bristol and Leeds sit at roughly half that (43). Investigate whether that reflects lower demand or an access gap.

7. Take no action on blood group.

Recovery rates are consistent across all eight groups (about 0.33 to 0.44), and each group has a comparable number of patients (13 to 17). The spread is not a small-sample artifact, and no group needs targeted intervention.

Conclusion

Riverside General Hospital is financially stable and clinically effective, with clear places to improve. Revenue of 318K rests on a healthy 76.2% collection rate, and three in four patients leave recovered or improving. Satisfaction of 3.9 out of 5 and a two-day average stay point to well-run care.

Three findings stand out.

First, **revenue is driven by value per visit, not patient volume.** Visits are flat at 80 per month, orthopaedics and radiology earn far more per encounter than everything else, and the 30% revenue-per-visit dip explains the February to March slump.

Second, **the collection gap is concentrated.** Radiology and neurology trail by 13 points while six departments perform well.

Third, **the operational strain is in unplanned care.** About 59% of visits arrive as Walk-In or Emergency, and those patients wait the longest.

The model we built on scalable star schema, so as more months of data are loaded, the same measures and visuals will continue to surface trends automatically, giving leadership on going, rather than one-time, view into hospital performance.

Thank you

